MTH 451/551
Lecture 8

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1 Lagrange Interpolation: Lagrange and Barycentric Forms

[Textbook reference: §3.3]

Consider the following **Lagrange interpolation problem:**

> Given $n+1$ distinct numbers $\{x_0, x_1, \dots, x_n\}$ (called **nodes**) and $n+1$ arbitrary values $\{y_0, y_1, \dots, y_n\}$, determine the *unique* polynomial $p \in \mathbb{P}_n$ such that
>
> $$p(x_j) = y_j$$
>
> for all $0 \leq j \leq n$.

The polynomial p is called the **Lagrange interpolating polynomial** or **Lagrange interpolant**. Note that this interpolation problem involves only *values*, and not derivatives. In the case that y_j are the values of a function at the nodes, $y_j = f(x_j)$, we say that p **interpolates** f **at the nodes**.

The Lagrange basis provides a convenient expression for the polynomial p,

$$p(x) = \sum_{j=0}^{n} y_j \ell_j(x).$$

This is called the **Lagrange form** of the interpolating polynomial. This form is very useful for analysis and theory, but other forms of the polynomial are better for practical computations.

- Each evaluation of $\ell_j(x)$ requires $\mathcal{O}(n)$ operations, and so the number of operations to evaluate $p(x)$ in this form is $\mathcal{O}(n^2)$. However, evaluating a polynomial of degree n should only take $\mathcal{O}(n)$ operations, and so this is very suboptimal!
- This form if *numerically unstable*, meaning that floating point rounding errors can destroy its accuracy.

- Adding a new data pair (x_{n+1}, y_{n+1}) requires recomputing everything from scratch.

We will now describe the so-called **Barycentric form**, which is designed to address these shortcomings. Define the **node polynomial** by

$$\ell(x) = \prod_{j=0}^{n}(x - x_j) = (x - x_0)(x - x_1)\cdots(x - x_n).$$

This is a polynomial of degree $n + 1$ that vanishes at all of the nodes. Define the **barycentric weights** by

$$\lambda_j = \frac{1}{\prod_{\substack{k=0 \\ k\neq j}}^{n}(x_j - x_k)}.$$

The Lagrange basis is related to the node polynomial and the barycentric weights by the relation

$$\ell_j(x) = \frac{\ell(x)}{x - x_j}\lambda_j.$$

Then, the interpolating polynomial can be written as

$$\begin{aligned} p(x) &= \sum_{j=0}^{n} y_j \ell_j(x) \\ &= \sum_{j=0}^{n} y_j \frac{\ell(x)}{x - x_j}\lambda_j \\ &= \ell(x) \sum_{j=0}^{n} \frac{y_j \lambda_j}{x - x_j} \end{aligned}$$

This is called the **first barycentric form:** $\boxed{p(x) = \ell(x)\sum_{j=0}^{n} \frac{y_j\lambda_j}{x-x_j}}$. Note that once the weights are computed ($\mathcal{O}(n^2)$ startup cost) and stored, every evaluation of $p(x)$ in this form requires only $\mathcal{O}(n)$ operations. *The weights are completely independent of the data values y_j.* Also, adding a new data pair requires updating the weights ($\mathcal{O}(n)$ operations total) and computing the new weight (also $\mathcal{O}(n)$). So the Lagrange interpolant in Barycentric form can be updated in a total of $\mathcal{O}(n)$ operations.

We will now derive a second barycentric form. We saw previously that $\sum_{j=0}^{n} \ell_j = 1$, and so

$$1 = \sum_{j=0}^{n} \ell_j = \sum_{j=0}^{n} \frac{\ell(x)}{x - x_j}\lambda_j = \ell(x) \sum_{j=0}^{n} \frac{\lambda_j}{x - x_j},$$

from which we conclude that

$$\sum_{j=0}^{n} \frac{\lambda_j}{x - x_j} = \frac{1}{\ell(x)}, \qquad \ell(x) = \frac{1}{\sum_{j=0}^{n} \frac{\lambda_j}{x-x_j}}.$$

Therefore, from the first barycentric form, we can derive the **second barycentric form**,

$$\boxed{p(x) = \frac{\displaystyle\sum_{j=0}^{n} \frac{y_j \lambda_j}{x - x_j}}{\displaystyle\sum_{j=0}^{n} \frac{\lambda_j}{x - x_j}}.}$$

Note that if all λ_j share a common factor c, i.e. $\lambda_j = c\omega_j$, then this factor appears in both the numerator and denominator, and can be cancelled, giving

$$p(x) = \frac{\displaystyle\sum_{j=0}^{n} \frac{y_j \omega_j}{x - x_j}}{\displaystyle\sum_{j=0}^{n} \frac{\omega_j}{x - x_j}}$$

for any such ω_j. In other words, the barycentric formula is invariant under multiplication of the weights by a constant.

1.1 Choice of nodes

The most obvious choice of nodes are the so-called *equispaced* or *equidistant* nodes: $x_j = a + jh$, where $h = (b - a)/n$. We explicitly compute the weights in this case,

$$\begin{aligned}
\lambda_j^{-1} &= \prod_{\substack{k=0 \\ k \neq j}}^{n} (x_j - x_k) = \prod_{\substack{k=0 \\ k \neq j}}^{n} (a + jh - a - kh) \\
&= \prod_{\substack{k=0 \\ k \neq j}}^{n} (jh - kh) = \prod_{\substack{k=0 \\ k \neq j}}^{n} h(j - k) = h^n \prod_{\substack{k=0 \\ k \neq j}}^{n} (j - k) \\
&= h^n j!(-1)^{n-j}(n - j)! \\
&= h^n (-1)^{n-j} \frac{n!}{\binom{n}{j}},
\end{aligned}$$

and so

$$\lambda_j = \binom{n}{j} \frac{(-1)^{n-j}}{h^n n!}.$$

These weights share a common factor of $1/(h^n n!)$ which can be cancelled, giving

$$\omega_j = (-1)^{n-j} \binom{n}{j}.$$

For example, with $n = 12$, the weights are

j	0	1	2	3	4	5	6	7	8	9	10	11	12
ω_j	1	-12	66	-220	495	-792	924	-792	495	-220	66	-12	1

Notice that towards the middle of the table, the values get larger in magnitude, and alternate in sign. This means that small changes in the data towards the middle of the interval could have large impacts on the value of the formula towards the endpoints. In other words, we expect polynomial interpolation in this case to be **unstable**: small changes in the data can have large changes in the resulting polynomial. Note that this is **not** specific to the barycentric form (it just reveals the problem to us in a clear way). Rather, this is a problem **inherent** in polynomial interpolation at equispaced nodes.

We now consider another node set, called the **Chebyshev** nodes. The Chebyshev nodes on the interval $[a, b]$, with width $h = b - a$ and midpoint $c = (b + a)/2$, are given by

$$x_j = c + (h/2)\cos(j\pi/n).$$

These are the points that are **equidistributed on the semicircle of radius** $h/2$ **centered at** c. It can be shown that the reciprocals of the barycentric weights are

$$\lambda_j^{-1} = 2^{1-n}\left(\frac{h}{2}\right)^n n\begin{cases}(-1)^j, & 1 \le j \le n-1,\\ 2(-1)^j, & j = 0, n.\end{cases}$$

Cancelling the common factor, we obtain the simplified weights

$$\omega_j = (-1)^j\delta_j, \qquad \delta_j = \begin{cases}1, & 1 \le j \le n-1,\\ 1/2, & j = 0, n.\end{cases}$$

For example, with $n = 12$, the weights are

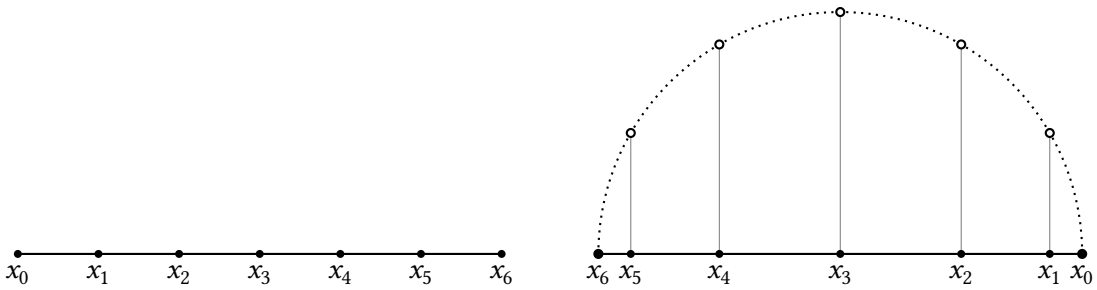
j	0	1	2	3	4	5	6	7	8	9	10	11	12
ω_j	1/2	-1	1	-1	1	-1	1	-1	1	-1	1	-1	1/2

In contrast to the equispaced case, all of these weights have the same magnitude (except for the two endpoints), so no part of the data is amplified relative to the rest.

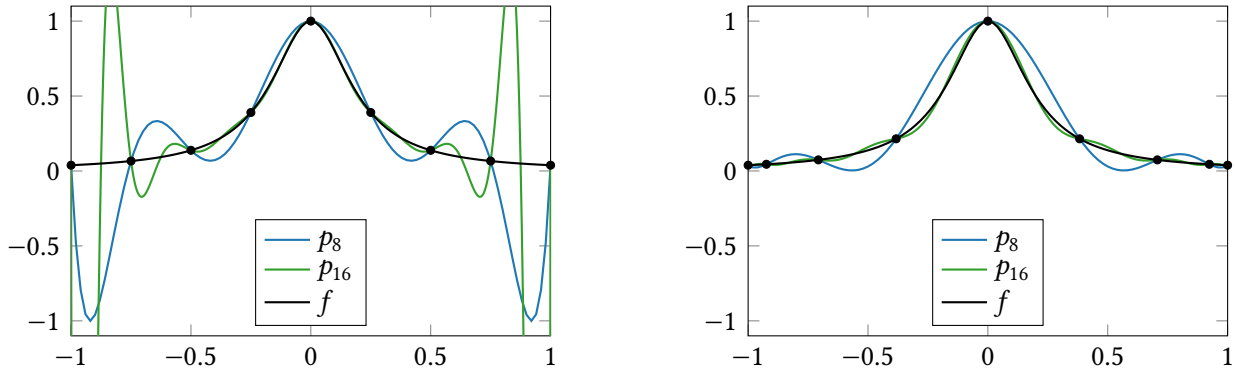
The Chebyshev nodes cluster near the endpoints of the interval. Near the endpoints, the spacing between nodes is

$$x_0 - x_1 = \frac{h}{2}\left(1 - \cos(\pi/n)\right) \approx \frac{\pi^2 h}{4n^2} = \mathcal{O}(1/n^2),$$

while near the midpoint, the spacing is approximately $\frac{\pi h}{2n} = \mathcal{O}(1/n)$, comparable to the equispaced spacing h/n. The picture below compares the equispaced nodes and the Chebyshev nodes for $n = 6$.



A classic example is the **Runge function** $f(x) = 1/(1 + 25x^2)$ on $[-1, 1]$. Compare its interpolant at $n+1$ equispaced nodes with its interpolant at the same number of Chebyshev nodes (on $[-1, 1]$, these are $x_j = \cos(j\pi/n)$).



The plots above show the Runge function f and its interpolating polynomials p_8 and p_{16} of degrees $n = 8$ and $n = 16$, at equispaced nodes (left) and at Chebyshev nodes (right); the nodes for $n = 8$ are marked. The equispaced interpolant has large errors near the endpoints, and these errors get *worse* as n increases. The Chebyshev interpolant, on the other hand, gets better as n increases.